

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/15]
Which property of a substance takes into account atmospheric pressure significantly?
A specific latent heat of fusion
B specific latent heat of vapourisation
C specific heat capacity of solid
D specific heat capacity of liquid

2. [AJC/H2/Prelims 2018/P1/12]
The first law of thermodynamics can be written as
$$q = \Delta U + w$$

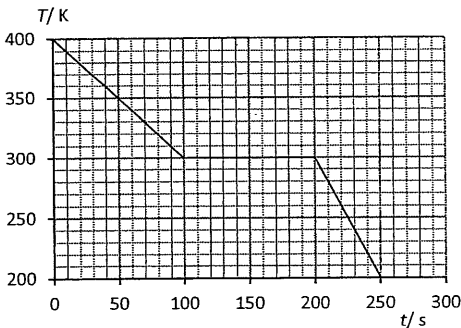
where ΔU is the increase in internal energy.
Which of the following correctly represent what q and w stands for?

	Q	W
A	Thermal energy supplied to the system	Work done on the system
B	Thermal energy supplied to the system	Work done by the system
C	Thermal energy removed from the system	Work done on the system
C	Thermal energy removed from the system	Work done by the system

3. [EJC/H2/Prelims 2018/P1/11]
A fixed amount of ideal gas undergoes the following changes:
Process 1: The gas is heated at constant volume.
Process 2: The gas is compressed at constant pressure.
How does the internal energy of the gas for each process change?

	Process 1	Process 2
A	increase	increase
B	increase	decrease
C	decrease	increase
C	decrease	decrease

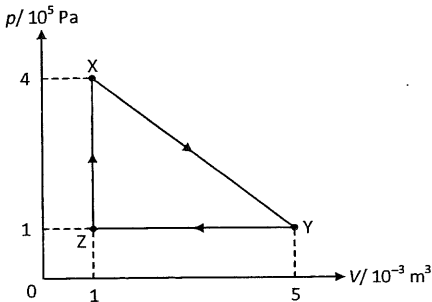
4. [HCI/H2/Prelims 2018/P1/17]
The graph shows the variation of temperature T against time t of a certain substance. Originally, it is in a liquid state at $t = 0$ s. Heat is removed from it at a constant rate until it becomes a solid.



Which of the following could be correct?

	Specific heat capacity of liquid/ $\text{J kg}^{-1} \text{K}^{-1}$	Specific heat capacity of solid/ $\text{J kg}^{-1} \text{K}^{-1}$
A	1500	3000
B	1800	900
C	2500	2500
D	4500	3000

5. [IJC/H2/Prelims 2018/P1/13 Modified]
A mass of an ideal gas of volume V , at pressure p , undergoes a cycle of changes as shown in the graph.



Which combination is correct?

	Work done by gas is negative	Heat is
A	Y to Z	added
B	Y to Z	removed
C	X to Y	added
D	X to Y	removed

6. [JJC/H2/Prelims 2018/P1/12]

A piston is pushed into a cylinder containing an ideal gas such that the pressure of the gas increases to 1.5 times its initial value while the volume decreases to half its original value.

Which one of the following statements is correct?

- A The average random kinetic energy of the gas molecules increases.
- B Work is done by the gas on its surroundings.
- C Heat is extracted from the gas.
- D There is no change in the internal energy of the gas.

7. [NJC/H2/Prelims 2018/P1/13]

A mole of monatomic ideal gas has pressure p_A and volume V_A initially. The gas undergoes a change to a final pressure p_B and volume V_B such that $V_B > V_A$.

Which of the following statements is true?

- A The heat supplied to the gas during the process can be determined using only the values p_A , V_A , p_B and V_B .
- B The change in the internal energy of the gas during the process can be determined using only the values p_A , V_A , p_B and V_B .
- C The work done by the gas during the process can be determined using only the values p_A , V_A , p_B and V_B .
- D None of the above.

8. [NYJC/H2/Prelims 2018/P1/14]

In a heating experiment, energy is supplied at a constant rate to a liquid in a beaker of negligible heat capacity. The temperature of the liquid rises by 4.0 K per minute just before it begins to boil. After 40 minutes, all the liquid has boiled away.

For this liquid, what is the ratio $\frac{\text{specific heat capacity}}{\text{specific latent heat of vapourisation}}$?

- A $\frac{1}{10} \text{ K}^{-1}$
- B $\frac{1}{40} \text{ K}^{-1}$
- C $\frac{1}{160} \text{ K}^{-1}$
- D $\frac{1}{640} \text{ K}^{-1}$

9. [PJC/H2/Prelims 2018/P1/9]

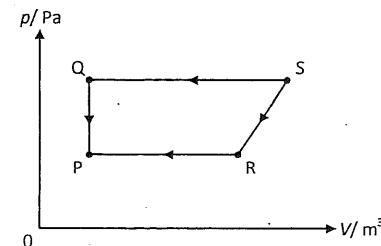
To cool down the electrical generator of a nuclear power plant, cold water enters the heat exchanger of the generator at 3 °C and leaves at 11 °C. The rate of heat removed by the water is $4.0 \times 10^{11} \text{ J per hour}$. The specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

What is the rate of water flow?

- A $\frac{4.0 \times 10^{11}}{4200 \times 8 \times 60} \text{ kg s}^{-1}$
- B $\frac{4.0 \times 10^{11} \times 60 \times 60}{4200 \times 8} \text{ kg s}^{-1}$
- C $\frac{4.0 \times 10^{11}}{4200 \times 8 \times 60} \text{ kg s}^{-1}$
- D $\frac{4.0 \times 10^{11} \times 60}{4200 \times 8} \text{ kg s}^{-1}$

10. [PJC/H2/Prelims 2018/P1/11]

A fixed mass of ideal gas undergoes changes in pressure and volume as shown below.



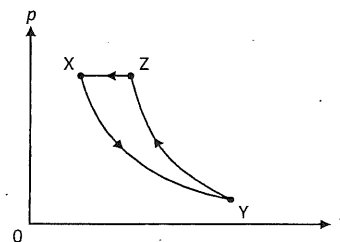
For $S \rightarrow Q \rightarrow P$, work done is 3 J and 8 J of heat is released by the gas. For $S \rightarrow R \rightarrow P$, work done is 2 J.

Which statement about the heat exchange for the change $S \rightarrow R \rightarrow P$ is correct?

- A 3 J of heat is released
- B 7 J of heat is released
- C 9 J of heat is released
- D 13 J of heat is released

11. [RVHS/H2/Prelims 2018/P1/9]

A fixed mass of an ideal gas undergoes the changes represented by XYZ below.



Which one of the following sets could describe this set of changes?

	XY	YZ	ZX
A	isothermal expansion	adiabatic compression	pressure reduction at constant volume
B	adiabatic expansion	isothermal compression	compression at constant pressure
C	isothermal expansion	adiabatic compression	compression at constant pressure
D	adiabatic expansion	isothermal compression	pressure reduction at constant volume

12. [VJC/H2/Prelims 2018/P1/10]

The specific heat capacity of a liquid is to be found using a continuous flow calorimeter. First, an input power of 10 W is used. When the input power is 18 W it is found that the liquid flow rate must be tripled to give the same temperature rise.

What is the rate of heat lost to the surroundings?

- A 3.3 W
- B 6.0 W
- C 6.7 W
- D 7.5 W

13. [IJC/H2/Prelims 2018/P1/12]

Two bodies X and Y are in good thermal contact and no net transfer of heat energy from one to the other is observed. Which one of the following statements is certainly correct?

- A The mean kinetic energies of their molecules are the same.
- B The total kinetic energies of their molecules are the same.
- C The materials have the same thermal conductivity.
- D Their internal energies are equal.

14. [MJC/H2/Prelims 2018/P1/15]

When a volatile liquid evaporates it cools down. What is the reason for this cooling?

- A All the molecules slow down.
- B Fast molecules leave the surface so the mean speed of those left behind is reduced.
- C Molecular collisions result in loss of kinetic energy of the molecules.
- D The molecules collide with one another less frequently

15. [NJC/H2/Prelims 2018/P1/12]

Body X and body Y have the same temperature. Body Y is in thermal equilibrium with body Z. The three bodies are of different materials. Body X has the largest mass and body Z has the least mass.

Which one of the following statements is correct?

- A X and Y have the same internal energy
- B Y and Z have the same specific heat capacity.
- C If X is placed in thermal contact with Z, there is no net transfer of heat.
- D If X is placed in thermal contact with Z, heat will be transferred from X to Z.

16. [SRJC/H2/Prelims 2018/P1/14]

Which of the following statements about the internal energy of a monatomic ideal gas is correct?

- A It will increase when heat is supplied to the gas.
- B It is proportional to the root-mean-square speed of the gas
- C It increases when the temperature of the gas increases.
- D It is dependent on the potential energy of the gas.

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P3/3]

- a. State what is meant by *internal energy* of a gas. [1]
- b. 1.2×10^{-5} mol of ideal monoatomic gas is held in a container which allows heat to be freely exchanged between the gas, a regenerator and the surroundings. The regenerator acts as a type of heat storage that allows the gas to transfer heat to and from the regenerator as the gas cycles through the 4 states. The gas is made to undergo the following thermodynamic cycle as shown in Fig. 1.1.

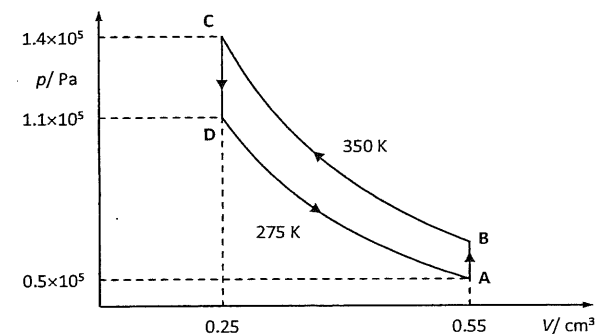


Fig. 1.1

A → B: the gas absorbs heat from the regenerator and its pressure increases.

B → C: the gas contracts isothermally at 350 K

C → D: the gas supplies heat to the regenerator and its pressure decreases

D → A: the gas expands isothermally at 275 K

i. Determine the pressure at B. [2]

ii. Show that the change in internal energy from C → D is -11.2×10^{-3} J. [1]

iii. Fig. 1.2 shows the energy changes for the transitions from A → B → C → D → A.

Complete the table. Show your working clearly. [3]

	Heat supplied to gas/ 10^{-3} J	work done on gas/ 10^{-3} J	increase in internal energy/ 10^{-3} J
A → B			
B → C		27.5	0
C → D			-11.2
D → A		-21.6	0

Fig. 1.2

iv. The above cycle can be used to draw heat from the surroundings by doing work on the gas. The coefficient of performance (COP) is a measure of how efficient this process is. The COP for such a system is defined as

$$Q_{in}/W_{net}$$

where Q_{in} is the heat absorbed by the gas from the surroundings and W_{net} is the net work done on the system for one cycle.

1. State which isothermal transition the gas absorbs heat energy from the surroundings. [1]

2. Hence, determine the COP for this cycle. [2]

[Total: 10]

2. [AJC/H2/Prelims 2018/P3/3]

- State what is meant by specific latent heat of vaporisation. [2]
- As water turns to steam, the intermolecular separation increases by ten times. The atmospheric pressure is 1.0×10^5 Pa, the density of water is 1000 kg m^{-3} and the specific latent heat of vaporization is 2300 J g^{-1} .
 - The water molecules experience increase in internal energy when the water boils. State what is meant by increase in internal energy. [1]
 - Using the first law of thermodynamics, determine the change in internal energy ΔU when 1.0 kg of water completely boils off to become steam. [4]
 - The latent heat of vaporization is not equal to the increase in internal energy. Explain why they are different. [1]

[Total: 8]

3. [CJC/H2/Prelims 2018/P2/3]

- Define *specific latent heat*. [1]
- A beaker containing a liquid is placed on a balance, as shown in Fig. 3.1.

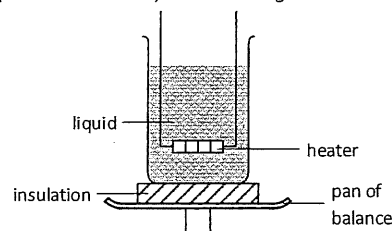


Fig. 3.1

A heater of power 120 W is immersed in the liquid. The heater is switched on and, when the liquid is boiling, balance readings M are taken at corresponding times t .

A graph of the variation with time t of the balance reading M is shown in Fig. 3.2.

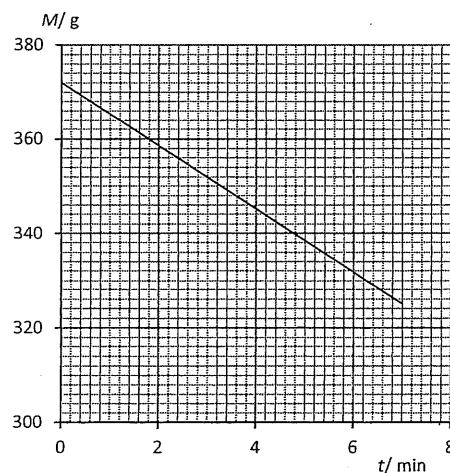


Fig. 3.2

- State the feature of Fig. 3.2 which suggests that the liquid is boiling at a steady rate. [1]
- Use data from Fig. 3.2 to determine a value for the specific latent heat of vaporisation l_v of the liquid. Explain your working. [3]
- State, with a reason, whether the experimental value determined in (b)(ii) is likely to be an overestimate or an underestimate of the expected value for the specific latent heat of vaporisation of the liquid. [2]

[Total: 7]

4. [CJC/H2/Prelims 2018/P3/2]

In a diesel engine, a fixed amount of gas undergoes a cycle of four stages. The cycle is shown in Fig. 4.1.

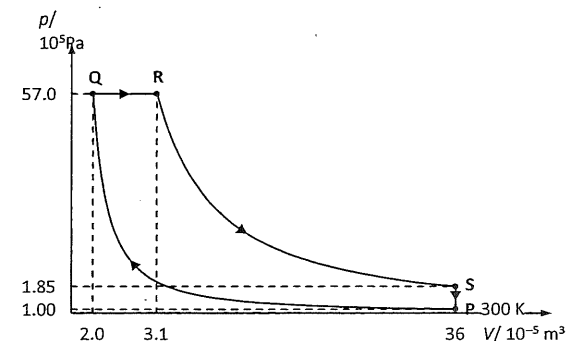


Fig. 4.1 (not to scale)

The four stages are

- $P \rightarrow Q$: compression with a rise in temperature and pressure,
 $Q \rightarrow R$: expansion at constant pressure while fuel is being burnt,
 $R \rightarrow S$: expansion with a drop in both temperature and pressure,
 $S \rightarrow P$: decrease in pressure at constant volume.

Some numerical values of temperature, pressure and volume are given on Fig. 4.1.

- Using Fig. 4.1, calculate the work done by the gas during the stages
 - $Q \rightarrow R$, [1]
 - $S \rightarrow P$. [1]
- Using your answers in (a), complete Fig. 4.2 for the four stages of the cycle. [4]

Stage of cycle	heat supplied to gas/ J	work done on gas/ J	increase in internal energy/ J
$P \rightarrow Q$	0	235	
$Q \rightarrow R$	246		
$R \rightarrow S$	0	-333	
$S \rightarrow P$			

Fig. 4.2

- Assuming that the gas is ideal, calculate the temperature of the gas at point Q. [2]

[Total: 8]

5. [DHS/H2/Prelims 2018/P3/5 Modified]

A cycle of changes in pressure, volume and temperature of gas inside a cylinder of a petrol engine is illustrated in Fig. 5.1. The gas is assumed to be ideal.

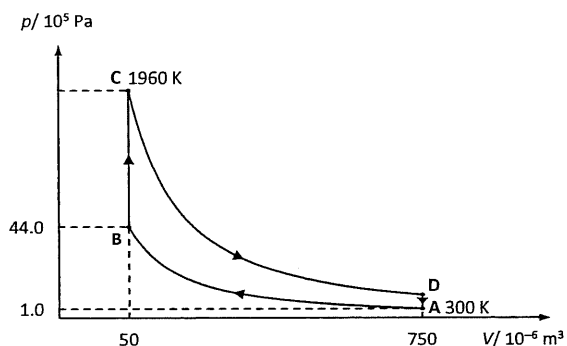


Fig. 5.1 (not to scale)

There are four stages in the cycle are shown in Fig. 5.2.

Stage	Description
A to B	Rapid compression of the gaseous petrol/air mixture with the temperature rising from 300 K at A and the pressure rising to 44.0×10^5 Pa at B.
B to C	The petrol/air mixture is exploded, resulting in an almost instant rise in pressure. At C the temperature has risen to 1960 K.
C to D	Rapid expansion and cooling of the hot gases.
D to A	Return to the starting point of the cycle.

Fig. 5.2

- a. i. State what is meant by an ideal gas. [2]
 ii. Use the values in Fig. 5.1 to determine the number of moles present in the gases in the cycle. [2]
 b. Complete the table in Fig. 5.3 showing the work done on the gas, the heat supplied to the gas and the increase in the internal energy of the gas, during the four stages of one cycle. [4]

Stage	work done on gas/J	heat supplied to gas/J	increase in internal energy of gas/J
A to B	+360	0	
B to C		+670	
C to D		0	-810
D to A			

Fig. 5.3

- c. Explain qualitatively how molecular movement causes the fall in temperature of the gas during the stage from C to D. [2]

[Total: 10]

6. [EJC/H2/Prelims 2018/P3/4]

- a. State the First Law of Thermodynamics. [1]
 b. 0.050 mol of an ideal gas is contained within an uninsulated cylinder with a movable piston as shown in Fig. 6.1.

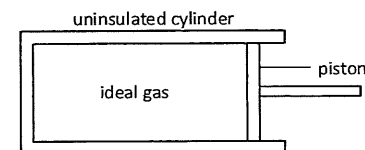


Fig. 6.1

The piston moves slowly outwards, resulting in the variation of pressure shown in Fig. 6.2.

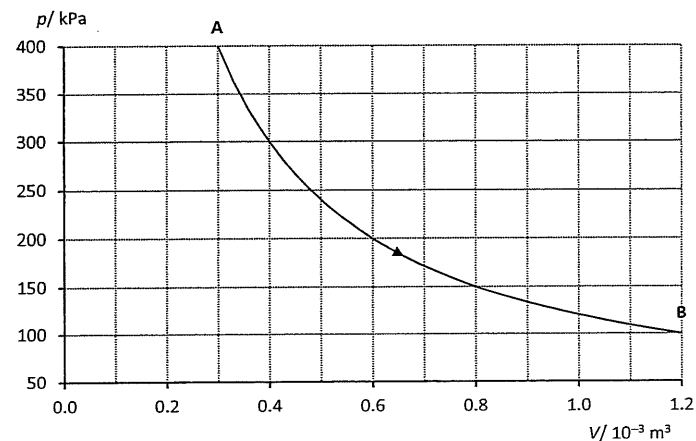


Fig. 6.2

- i. The process A to B occurs along an isotherm. Determine the temperature along this isotherm. [2]
 ii. Calculate the total kinetic energy as a result of the random motion of the gas molecules in the cylinder. [1]
 iii. Estimate the amount of work done by gas as it expands from A to B. [1]
 iv. A student states that no heat flows into or out of the gas during the process A to B as the temperature of the gas did not change. State and explain the validity of his statement. [3]
 c. In another set up with the same starting point as the one shown in Fig. 6.1, the piston is moved very quickly outwards expanding the gas to a volume of $1.2 \times 10^{-3} \text{ m}^3$. As a result, the temperature of the gas decreases to 144 K. On Fig. 6.2, sketch the variation with volume of the pressure of the gas. [2]

[Total: 10]

7. [H1/H2/Prelims 2018/P3/2 Modified]

- Explain what is meant by the internal energy of a system.
- Dry air is enclosed in an air-tight cylinder fitted with a piston, as shown in Fig. 7.1.

[2]

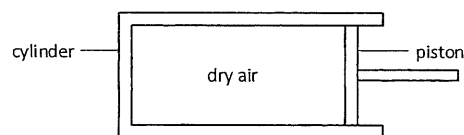


Fig. 7.1

The piston moves to compress the air and the variation with volume V of the pressure p of the air during the process from A to B is shown in Fig. 7.2.

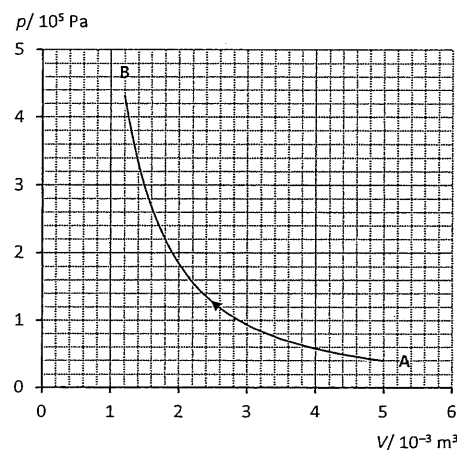


Fig. 7.2

It may be assumed that the dry air behaves as an monatomic ideal gas.

- Calculate the internal energy of the dry air just before the start of process from A to B.
- The dry air then goes through two more processes.
 Process 2: The air is cooled while keeping the piston at the same position.
 Process 3: The air then expands, while kept at constant temperature, to return to its original state.
 - Calculate the pressure of the air at the end of the process 2.
 - On Fig. 7.2, draw accurately and label on the p - V graph of the two processes 2 and 3.

[Total: 9]

8. [IJC/H2/Prelims 2018/P2/2]

Fig. 7.1 shows a simplified heat engine which consists of an insulated cylinder fitted with a perfectly fitting frictionless piston. The cylinder contains a fixed mass of an ideal gas and a heater. When heat is supplied to the gas, it can expand and does work by pushing the piston.

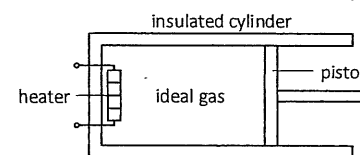


Fig. 7.1

- Explain in terms of the motion of the molecules in the gas, how work can be done by the heated gas to move the piston.
- The efficiency of a heat engine is given by the following equation

$$\text{efficiency} = \frac{\text{useful mechanical work done}}{\text{input heat supplied}}$$

- In practice, attaining 100% efficiency is impossible. Suggest two possible reasons.
- Two experiments are performed on the heat engine in Fig. 7.1.

Experiment 1

The heater provides 150 J of thermal energy with the piston held in a fixed position. The temperature rise of the gas is 30 K.

Experiment 2

The heater again supplies 150 J of thermal energy with the piston free to move so that the gas expands at a constant pressure of $1.0 \times 10^5 \text{ Pa}$ and does some useful work W . In this case, the temperature rise is 18 K.

- Explain why there is a difference in the temperature rise in each case.
- In the first experiment, calculate the gain in internal energy of the gas.
- In the second experiment, calculate the gain in internal energy of the gas.
- In the second experiment, calculate the work done W by the expanding gas.

[Total: 10]

9. [MI/H2/Prelims 2018/P3/3 Part]

- A student designed an air-filled "load-lifting" system where the air could be uniformly heated by suitably located electrical heaters in the enclosure of the system. All sides of the enclosure have good thermal insulation and the top surface is movable vertically.

The volume of air inside the enclosure at temperature of 300 K and pressure of 101.0 kPa is 2.000 m^3 .

When a load of mass 500 kg is placed on the top of the movable surface of the system, a depression of 6.8 cm of the top surface is observed and the volume of air in the enclosure is reduced to 1.933 m^3 .

- State the first law of thermodynamics.
- The air may be considered as an ideal gas. Determine
 - the change in internal energy of the air in the system after the load is placed on top of the movable surface.
 - the new pressure of air in the system.

[Total: 8]

10. [MJC/H2/Prelims 2018/P2/4]

- a. Fig. 10.1 shows a piece of heated metal, of mass 50 g, immersed in 130 g of water contained in a calorimeter.

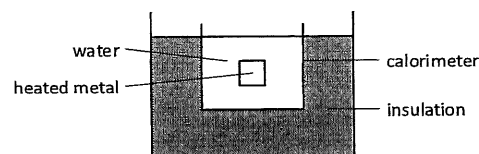


Fig. 10.1

The water into which the metal has been placed is stirred until it reaches a steady temperature.

The following data are available:

heat capacity of metal	82.7 J K^{-1}
specific heat capacity of the water	$4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
heat capacity of the calorimeter	54.6 J K^{-1}
initial temperature of the water	25°C
final temperature of the water	90°C

Use the data to calculate the initial temperature of the heated metal and state an assumption made for your calculation. [3]

- b. The gas in the cylinder of a diesel engine can be considered to undergo a cycle of changes of pressure, volume and temperature. One such cycle, for an ideal gas, is shown in Fig. 10.2. Processes A to B and C to D take place without heat exchange with the surroundings.

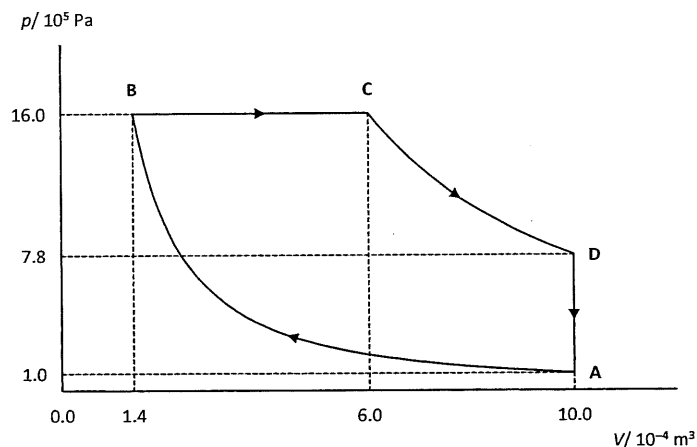


Fig. 10.2

Complete the table below.

Process	Heat supplied to gas/ J	Work done on gas/ J	Increase in internal energy of gas/ J
A to B		300	
B to C	2580		
C to D		-440	
D to A	-1700		

[4]

- c. A fixed mass of ideal gas is heated from temperature T_1 to T_2 at constant volume. Explain why a greater amount of heat is required to heat the same mass of ideal gas from T_1 to T_2 at constant pressure. [3]

[Total: 10]

11. [NJC/H2/Prelims 2018/P3/6]

- a. A mass of 24 g of ice at -15°C is taken from a freezer and placed in an insulated container with 200 g of water at 28°C . Some data for the ice and the water are given in the table below.

	specific heat capacity/ $\text{J kg}^{-1} \text{ K}^{-1}$	specific latent heat of fusion/ J kg^{-1}
ice	2.1×10^3	3.3×10^5
water	4.2×10^3	-

Calculate the final temperature of the water in the container. [2]

- b. Some air, assumed to be an ideal gas, is contained within a cylinder. Initially, the volume of gas is 540 cm^3 , its pressure is $1.1 \times 10^5 \text{ Pa}$ and its temperature is 27°C . The air is compressed to 30 cm^3 suddenly such that no heat enters or leaves the gas. The pressure of the gas is $6.5 \times 10^6 \text{ Pa}$ immediately after the compression.

- Determine the temperature of the gas immediately after the compression. [2]
- Explain, using the first law of thermodynamics, why the temperature of the air changes during the compression. [3]

- c. The first law of thermodynamics can be expressed in the form

$$\Delta U = Q + W$$

where

ΔU is the increase of the internal energy of the system,

Q is the thermal energy supplied to the system, and

W is the work done on the system.

The table below shows different processes. Complete the table with symbols '+' for an increase, '-' for a decrease and the symbol '0' for no change, for the respective quantities U , Q and W . [3]

	U	Q	W
Compression of an ideal gas at constant temperature			
Heating of a solid with no expansion			
Melting of ice at 0°C into water at 0°C Note: ice is less dense than water			

[Total: 10]

12. [NYJC/H2/Prelims 2018/P3/5 Part Modified]

- d. i. Calculate the internal energy of a monatomic ideal gas of volume $3.4 \times 10^{-4} \text{ m}^3$ and pressure 100 kPa. [1]
- ii. Calculate the increase in internal energy when the gas expands at constant pressure to a volume of $8.8 \times 10^{-4} \text{ m}^3$. [2]
- iii. Determine the heat supplied to the gas so as to enable the expansion at constant pressure to take place. [2]

[Total: 5]

13. [RVHS/H2/Prelims 2018/P3/2 Modified]

- a. The first law of thermodynamics states that the increase in internal energy of a closed system is the sum of the heat supplied and work done on the system.

Explain what is meant by

- i. internal energy, [1]
- ii. closed system, [1]
- iii. heat. [1]
- b. i. An ideal gas occupies a volume V at a pressure p and a temperature of T inside an insulated container as shown in Fig. 13.1. It is held in place by a frictionless piston. The other section of the container is a vacuum.
- Using the first law of thermodynamics, explain why the temperature of the gas remains constant when the piston is no longer held in place and the gas is allowed to expand to occupy the entire volume of the container. [3]

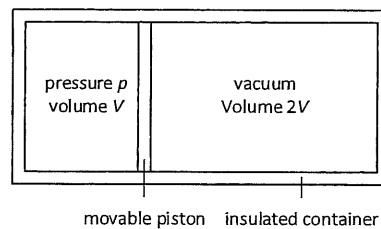


Fig. 13.1

- ii. Hence or otherwise, determine the ratio of the initial pressure to the final pressure of the gas. [2]
- c. In 1816, Robert Stirling patented the Stirling engine which operates by cyclic compression and expansion of air or other gas at different temperatures, such that there is a net conversion of thermal energy to mechanical work.

Figure 13.2 shows the cycle of changes that a gas undergoes in the Stirling engine.

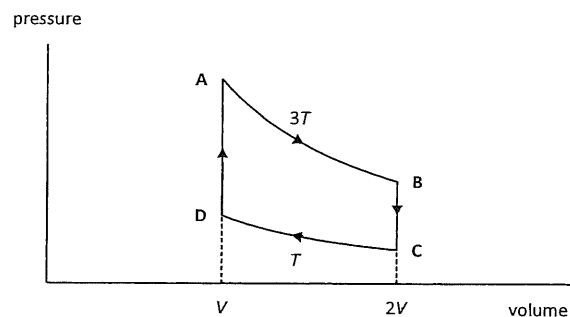


Fig. 13.2

Fig. 13.3 shows the processes that are involved in the cycle of changes.

process	description	heat supplied
A → B	increase of volume at constant temperature	
B → C	decrease of pressure at constant volume	
C → D	decrease of volume at constant temperature	
D → A	increase of pressure at constant volume	

Fig. 13.3

- i. Complete the table in Fig. 13.3 by writing 'to gas' or 'by gas'. [2]
- ii. Determine, in terms of n , R and T , the change in internal energy for n mol of a monatomic ideal gas for the change D → A. [1]
- iii. The work done on the gas for a change in volume at constant temperature can be given by the expression

$$W = nRT \ln \frac{V_i}{V_f}$$

where V_i and V_f are the initial and final volume of the gas.Hence or otherwise, derive an expression for the net heat transferred to the gas during one cycle in terms of n , R and T . Explain your working clearly. [3]

- iv. The efficiency of the Stirling engine shown is determined by the net work done by the system divided by the heat transferred to the system. Using your working in c.i. to iii., determine the efficiency. [2]

[Total: 16]

14. [SRJC/H2/Prelims 2018/P3/2 Modified]

A fixed mass of an ideal monatomic gas undergoes a cycle of changes ABCA, as shown in Fig. 14.1.

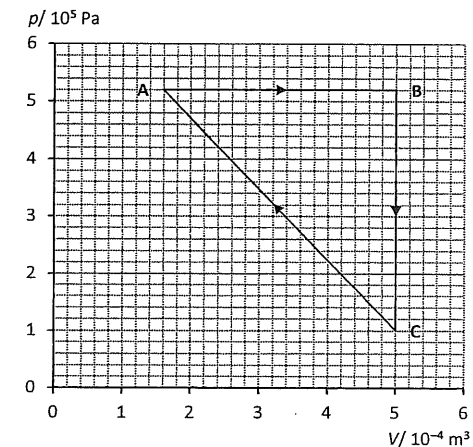


Fig. 14.1

- a. During the change from B to C, the internal energy of the gas decreases by 315 J.
By considering molecular energy, state and explain qualitatively the change, if any, in the temperature of the gas. [2]
- b. During the change from A to B, the heat supplied to the gas is 442 J.
Fig. 2.2 is a table of energy changes during one cycle. Complete Fig. 14.2. [3]

process	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
A to B	442		
B to C			-315
C to A			

Fig. 14.2

- c. Calculate the root-mean-square speed of the gas atoms at point B if the root-mean-square speed at point A is 350 m s^{-1} . [2]

[Total: 7]

15. [TPJC/H2/Prelims 2018/P2/5]

- a. A thermometer and an electrical heater are inserted into holes in an aluminium block of mass 960 g, as shown in Fig. 15.1.

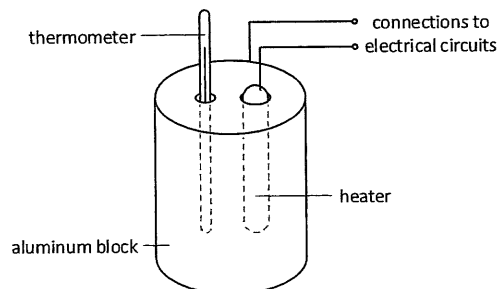


Fig. 15.1

The power rating of the heater is 54 W. The heater is switched on and readings of the temperature of the block are taken at regular time intervals. The variation with time t of the temperature θ of the block is shown in Fig. 15.2.

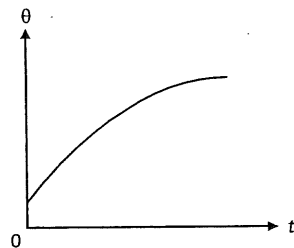


Fig. 15.2

- i. Suggest why the rate of rise of temperature of the block decreases to zero. [1]
- ii. When the block reaches a constant temperature, the heater is switched off and then further temperature readings are taken. The maximum rate of fall of temperature is found to be around $3.7 \text{ }^{\circ}\text{C min}^{-1}$.
Estimate the heat capacity of the aluminium block. [2]
- b. Fig. 15.3 shows an insulated cylinder fitted with a perfectly fitting piston and a heater. There is negligible friction between the cylinder and the piston. The cylinder contains a fixed mass of an ideal gas.

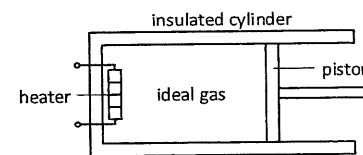


Fig. 15.3

The following two experiments are performed.

Experiment 1 The heater provides 150 J of energy with the piston fixed in position. The temperature rise of the gas is found to be 29 K.

Experiment 2 The heater supplies 150 J of energy with the piston free to move so that the gas expands at constant pressure. The temperature rise of the gas is now 18 K.

- i. Explain why the temperature rise is different in the two experiments. [2]
- ii. Fill up the table below to show the values of each of the quantities for experiments 1 and 2.
Present your workings using the space provided below. [4]

	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
Experiment 1			
Experiment 2			

[Total: 9]

16. [YJC/H2/Prelims 2018/P2/4]

- a. State the First Law of Thermodynamics in words. [1]
- b. A gas in an engine undergoes a cyclic process ABCD, as shown in Fig. 16.1.

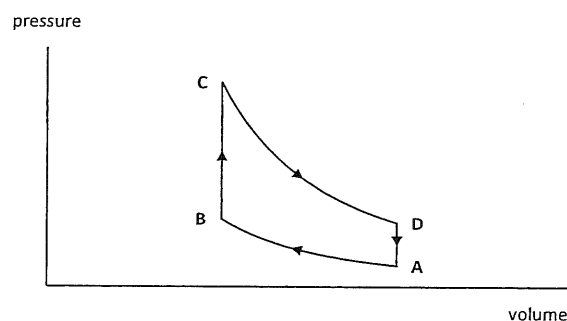


Fig. 16.1

AB and CD are adiabatic processes, while BC and DA are isochoric processes.

- i. For process AB, the work done is 10 J.
Calculate the change in internal energy of the gas. State whether it is an increase or decrease. [2]
- For process BC, 70 J of heat is supplied to the gas, while for process CD, 25 J of work is done.
- ii. Calculate the heat flow during process DA. State whether it is into or out of the gas. [3]
- iii. Calculate the efficiency of the engine. [2]
- iv. The engine executes 3000 cycles per minute. Calculate its useful power output. [2]

[Total: 10]

17. [YJC/H2/Prelims 2018/P2/3 Part]

- b. A refrigerator door is opened and room temperature air (25 °C) filled the 1.5 m³ compartment of the refrigerator. A 2.0 kg chicken, also at room temperature, is placed in the refrigerator and the door is closed.
If the power rating of the refrigerator is 200 W, with an efficiency of 0.80, calculate the time required to cool the chicken and air in the refrigerator to thermal equilibrium at a temperature of 4 °C.
(Density of air = 1.25 kg m⁻³, specific heat capacity of air = 1.02 × 10³ J kg⁻¹ K⁻¹, specific heat capacity of chicken = 3.48 × 10³ J kg⁻¹ K⁻¹) [3]

[Total: 3]

18. [YJC/H2/Prelims 2018/P3/4 Modified]

- a. State the first law of thermodynamics. [1]
- b. A diesel engine containing an ideal gas at a pressure of 1.0×10^5 Pa, temperature of 300 K and volume of 1.0×10^{-3} m³ undergoes a cyclic process in the sequence as follows:
A to B: the gas is compressed adiabatically (i.e., no heat transfer) to a pressure of 16.0×10^5 Pa, temperature of 660 K and volume of 1.4×10^{-4} m³.
B to C: the gas is expanded at constant pressure to a volume of 6.0×10^{-4} m³.
C to D: the gas continued to expand adiabatically to a pressure of 7.8×10^5 Pa to its original volume.
D to A: the gas is cooled at constant volume to its original temperature and pressure.
- i. Illustrate these changes on the p-V diagram in Fig. 18.1. [4]



Fig. 18.1

- ii. Fig. 18.2 shows the increase in internal energy of the gas, heat supplied to the gas and work done on the gas during the cyclic process. Complete Fig. 18.2. [3]

Process	Heat supplied to gas/ J	Work done on gas/ J	Increase in internal energy of gas/ J
A to B		300	
B to C	2580		
C to D		-440	
D to A	-1700		

Fig. 18.2

- iii. Assuming that the efficiency of the diesel engine is defined as ratio of net work done by the gas in one complete cycle to heat supplied to gas during process B to C, calculate the efficiency of this engine. [2]

[Total: 10]

19. [JJC/H2/Prelims 2018/P3/8 Part]

- b. i. State what is meant by the *internal energy* of a system. [1]
- ii. In Fig. 19.1 below, place a tick (✓) against those changes where the internal energy of a system is increasing. [2]

Water freezing at constant temperature	
A stone falling under gravity in a vacuum	
Water evaporating at constant temperature	
Stretching a wire at constant temperature.	

Fig. 19.1

- iii. A cylinder of gas at constant volume is placed under the Sun so that its temperature rises. Explain the changes to the internal energy of the gas. [2]
- iv. The volume occupied by 1.00 mol of water at 100 °C is $1.87 \times 10^{-5} \text{ m}^3$. When the water is vaporised at an atmospheric pressure of $1.03 \times 10^5 \text{ Pa}$, the water vapour has a volume of $2.96 \times 10^{-2} \text{ m}^3$. The latent heat required to vaporise 1.00 mol of water at 100 °C and $1.03 \times 10^5 \text{ Pa}$ is $4.05 \times 10^4 \text{ J}$. Determine, for this change of state, [2]
- the work done on the system, [2]
 - the increase in internal energy of the system. [2]
- c. i. A student states wrongly that temperature measures the amount of thermal energy in a body. State and explain two observations that show why the statement is incorrect. [2]
- ii. Some crushed ice at 0 °C is placed in a funnel together with an electric heater, as shown in Fig. 19.2 below.

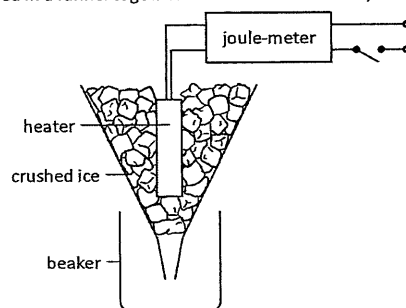


Fig. 19.2

The mass of water collected in the beaker in a measured interval of time is determined with the heater switched off. The mass is then found with the heater switched on. The energy supplied to the heater is also measured. For both measurements of the mass, water is not collected until melting occurs at a constant rate. The data shown in Fig. 19.3 are obtained.

	mass of water/ g	energy supplied to heater/ kJ	time interval/ min
heater switched off	8.3	0	5.0
heater switched on	64.7	18	5.0

Fig. 19.3

1. State why it is necessary to determine the mass of water with the heater switched off. [1]

2. Suggest how it can be determined that the ice is melting at a constant rate. [1]
3. Calculate a value for the specific latent heat of fusion of ice. [2]

[Total: 15]

